

Worksheet 5

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1. Aim/Overview of the practical:

Interface an Arduino Uno with Temp Sensor to calculate the temperature in Fahrenheit and print on a Serial monitor and on LCD. Follow the mentioned cases after the calculation of distance.

- If the temp is > 70 -degree F green light is in ON condition.
- If the temp is < 70 -degree F and temp is > 30 -degree F yellow light is in ON condition and the green light remains Off
- If the temp is < 30 degree F turn on the buzzer and the lights will remain OFF.

2. Apparatus (For applied/experimental sciences/materials based labs):

Hardware Requirements

- Arduino Uno
- LM35 Temperature sensor
- 16×2 LCD
- Green LED
- Yellow LED
- Buzzer
- Resistors (220Ω for LEDs)
- Breadboard and jumper wires

Software requirements

- Arduino IDE
- Tinkercad Circuits

3. Circuit Diagram (TinkerCad):

Code:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
```



```
#define LM35_PIN A0
#define GREEN_LED 7
#define YELLOW_LED 8
#define BUZZER 9

LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  Serial.begin(9600);

  pinMode(GREEN_LED, OUTPUT);
  pinMode(YELLOW_LED, OUTPUT);
  pinMode(BUZZER, OUTPUT);

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0, 0);
  lcd.print("LM35 Temp Sensor");
  delay(2000);
  lcd.clear();
}

void loop() {

  // Read LM35
  int sensorValue = analogRead(LM35_PIN);

  // Convert analog value to voltage
  float voltage = sensorValue * (5.0 / 1023.0);

  // Convert voltage to Celsius
  float tempC = voltage * 100.0;

  // Convert Celsius to Fahrenheit
  float tempF = (tempC * 9.0 / 5.0) + 32.0;

  // Turn everything OFF first
  digitalWrite(GREEN_LED, LOW);
  digitalWrite(YELLOW_LED, LOW);
  digitalWrite(BUZZER, LOW);

  // Temperature conditions
  if (tempF > 70) {
    digitalWrite(GREEN_LED, HIGH);
  }
  else if (tempF > 30 && tempF <= 70) {
    digitalWrite(YELLOW_LED, HIGH);
  }
  else if (tempF < 30) {
    digitalWrite(BUZZER, HIGH);
  }
}
```

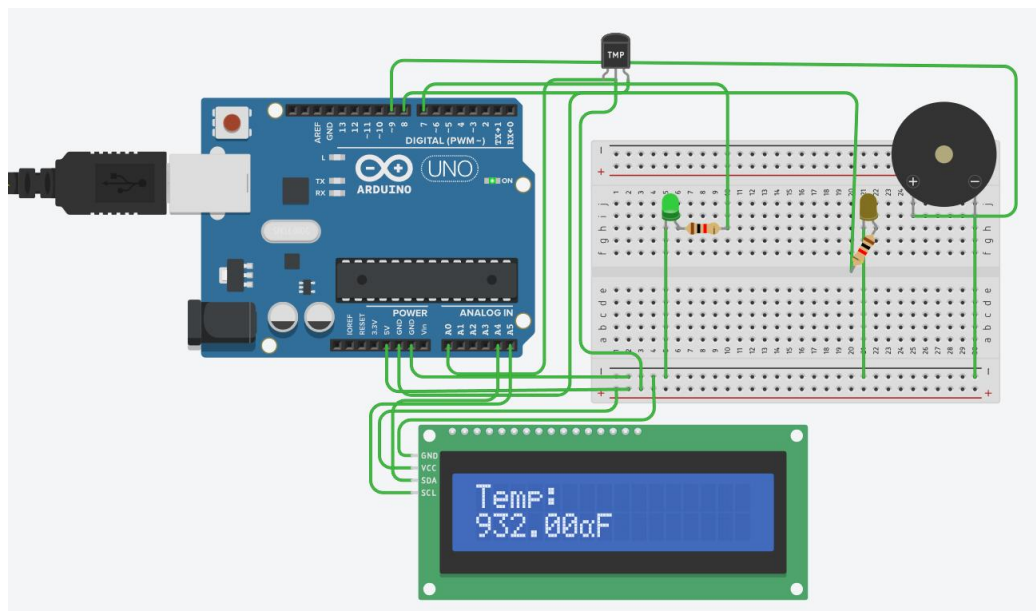
```
// Serial Monitor
Serial.print("Temperature: ");
Serial.print(tempF);
Serial.println(" F");

// LCD
lcd.clear();

lcd.setCursor(0, 0);
lcd.print("Temp:");

lcd.setCursor(0, 1);
lcd.print(tempF);
lcd.print((char)223);
lcd.print("F");

delay(1000);
}
```



5. Learning outcomes (What I have learnt):

1. Understand LM35 temperature sensor.
2. Calculate temperature in Fahrenheit.
3. Display temperature on LCD and Serial Monitor.
4. Control LEDs and buzzer based on temperature.